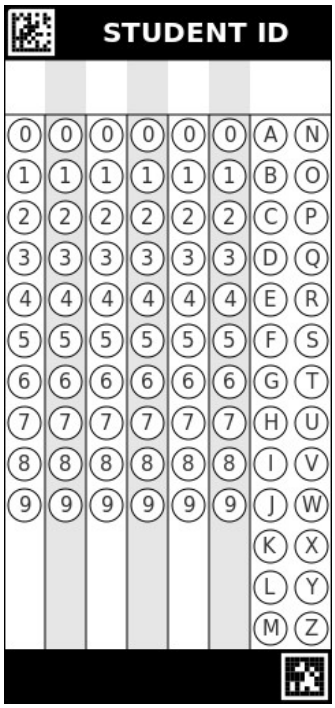
**HWA CHONG INSTITUTION**
**Preliminary Examination**
**Higher 2**

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# PHYSICS

**9478 / 03**

**Paper 3 Longer Structured Questions**

**18 September 2026**

**2 hours**

Candidates answer on the Question Paper.
No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

- Write your **name**, **CT group, centre number** and **index number** on all the work you hand in.
- **Write** and **shade** your Student ID using a 2B pencil. Your Student ID is xxxxxxx, where your HCI email is xxxxxxx@student.hci.edu.sg
- Use a dark blue or black pen. You may use a pencil for any diagrams or graphs.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen, correction fluid or tape.
- Do **not** remove any staple; Do **not** detach any pages of this question booklet.
- You may use an approved calculator.

**INFORMATION**

**Section A**
Answer all questions.
**Section B**
Answer one question only.

You are advised to spend one and a half hours on Section A and half an hour on Section B.

The number of marks for each question or part question is shown in brackets [ ].

| For Examiner's Use | | |
|---|---|---|
| **Section A** | | |
| **1** | | **5** |
| **2** | | **8** |
| **3** | | **8** |
| **4** | | **6** |
| **5** | | **8** |
| **6** | | **7** |
| **7** | | **6** |
| **8** | | **7** |
| **Section B** | | |
| **9** | | **20** |
| **10** | | **20** |
| **Deductions** | | |
| **P3** | | **75** |

**Data**

| | |
|---|---|
| speed of light in free space | $c = 3.00 \times 10^{8}$ m s$^{-1}$ |
| permeability of free space | $\mu_o = 4\pi \times 10^{-7}$ H m$^{-1}$ |
| permittivity of free space | $\varepsilon_o = 8.85 \times 10^{-12}$ F m$^{-1}$ |
| | $(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}$ m F$^{-1})$ |
| elementary charge | $e = 1.60 \times 10^{-19}$ C |
| Planck constant | $h = 6.63 \times 10^{-34}$ J s |
| unified atomic mass constant | $u = 1.66 \times 10^{-27}$ kg |
| rest mass of electron | $m_e = 9.11 \times 10^{-31}$ kg |
| rest mass of proton | $m_p = 1.67 \times 10^{-27}$ kg |
| molar gas constant | $R = 8.31$ J K$^{-1}$ mol$^{-1}$ |
| Avogadro constant | $N_A = 6.02 \times 10^{23}$ mol$^{-1}$ |
| Boltzmann constant | $k = 1.38 \times 10^{-23}$ J K$^{-1}$ |
| gravitational constant | $G = 6.67 \times 10^{-11}$ N m$^2$ kg$^{-2}$ |
| acceleration of free fall | $g = 9.81$ m s$^{-2}$ |

**Formulae**

| | |
|---|---|
| uniformly accelerated motion | $s = ut + \frac{1}{2}at^2$ |
| | $v^2 = u^2 + 2as$ |
| work done on/ by a gas | $W = p\Delta V$ |
| pressure | $p = \frac{F}{A}$ |
| gravitational potential | $\phi = -\frac{GM}{r}$ |
| temperature | $T / K = T / °C + 273.15$ |
| pressure of an ideal gas | $p = \frac{1}{3}\frac{Nm}{V} < c^2 >$ |
| mean translational kinetic energy of an ideal gas particle | $E = \frac{3}{2}kT$ |

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| | |
|---|---|
| displacement of particle in s.h.m. | $x = x_0 \sin \omega t$ |
| velocity of particle in s.h.m. | $v = v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$ |
| electric current | $I = nAvq$ |
| resistors in series | $R = R_1 + R_2 + \ldots$ |
| resistors in parallel | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$ |
| capacitors in series | $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$ |
| capacitors in parallel | $C = C_1 + C_2 + \ldots$ |
| energy in a capacitor | $U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$ |
| charging a capacitor | $Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$ |
| discharging a capacitor | $Q = Q_0 e^{-\frac{t}{\tau}}$ |
| RC time constant | $\tau = RC$ |
| electric potential | $V = \frac{Q}{4\pi\varepsilon_0 r}$ |
| alternating current / voltage | $x = x_0 \sin \omega t$ |
| magnetic flux density due to a long straight wire | $B = \frac{\mu_0 I}{2\pi d}$ |
| magnetic flux density due to a flat circular coil | $B = \frac{\mu_0 NI}{2r}$ |
| magnetic flux density due to a long solenoid | $B = \mu_0 nI$ |
| energy states for quantum particle in a box | $E_n = \frac{h^2}{8mL^2}n^2$ |
| radioactive decay | $x = x_0 e^{-\lambda t}$ |
| radioactive decay constant | $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$ |

## Section A

Answer **all** the questions in the spaces provided.

**1** A uniform rod AB of weight 12 N is freely hinged to a wall at A. It is held by a force *F* acting from B at an angle of 35° to the vertical as shown in Figure 1.1.

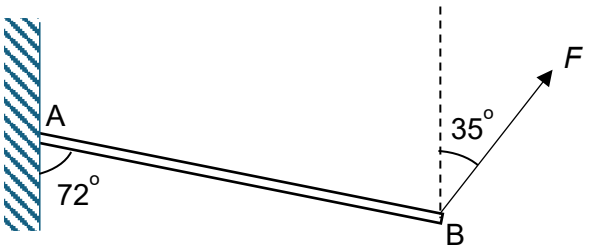



**Figure 1.1**

**(a)** Show that the force *F* is 6.0 N.

[1]

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**(b) (i)** Draw on Figure 1.1 the force $R$ acting on the rod at A.

[1]

**(ii)** Determine the magnitude of $R$.

$R$ = ..... N [3]

[Total: 5]

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**2** This question is about artificial satellites that orbit the Earth. You may treat the Earth as a perfect sphere with the centre of gravity at its centre. All other celestial bodies and their effects need not be considered.

The radius of the Earth is 6370 km and the mass of the Earth is $5.97 \times 10^{24}$ kg.

**(a) (i)** Determine the orbital period of a geostationary satellite in seconds.

period = ..... s [1]

**(ii)** Determine the orbital radius of a geostationary satellite to the centre of the Earth.

radius = ..... m [2]

**(b)** The Molniya orbit (Figure 2.1) is a highly elliptical orbital path designed to provide continuous communication coverage to polar regions by positioning its furthest point directly over the North Pole.

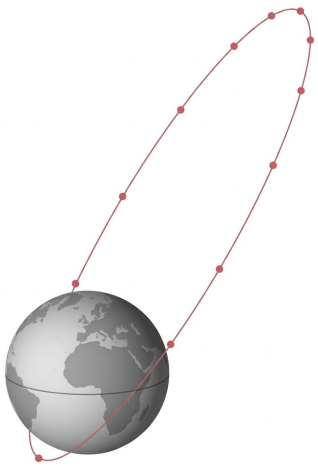


**Figure 2.1**

A satellite of mass 1500 kg is placed into this orbit. The closest distance the satellite approaches the Earth's centre is 6870 km while its furthest distance from the Earth's centre is 46200 km.

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**(i)** Calculate the change in gravitational potential experienced by the satellite as the satellite moves from the closest to the furthest points to Earth.

change in gravitational potential = ..... J $kg^{-1}$ [2]

**(ii)** Calculate the change in kinetic energy of the satellite as the satellite moves from the closest to the furthest points to Earth.

change in kinetic energy = ..... J [2]

**(iii)** With reference to your answer in **(b)(ii)**, explain why designing the orbit to have its furthest point over the pole makes it highly effective for continuous communications to the polar region.

.....

.....

..... [1]

[Total: 8]

**3** The needle-carrier of a sewing machine is constrained to move in a vertical line by fixed guides as shown in Figure 3.1. The simple harmonic motion of the needle-carrier is produced by a peg attached to the rim of a rotating disc which moves in a circle and engages the slot of the needle-carrier.

Figure 3.2 shows the front view of the positions of the peg and needle-carrier at intervals of a quarter period.

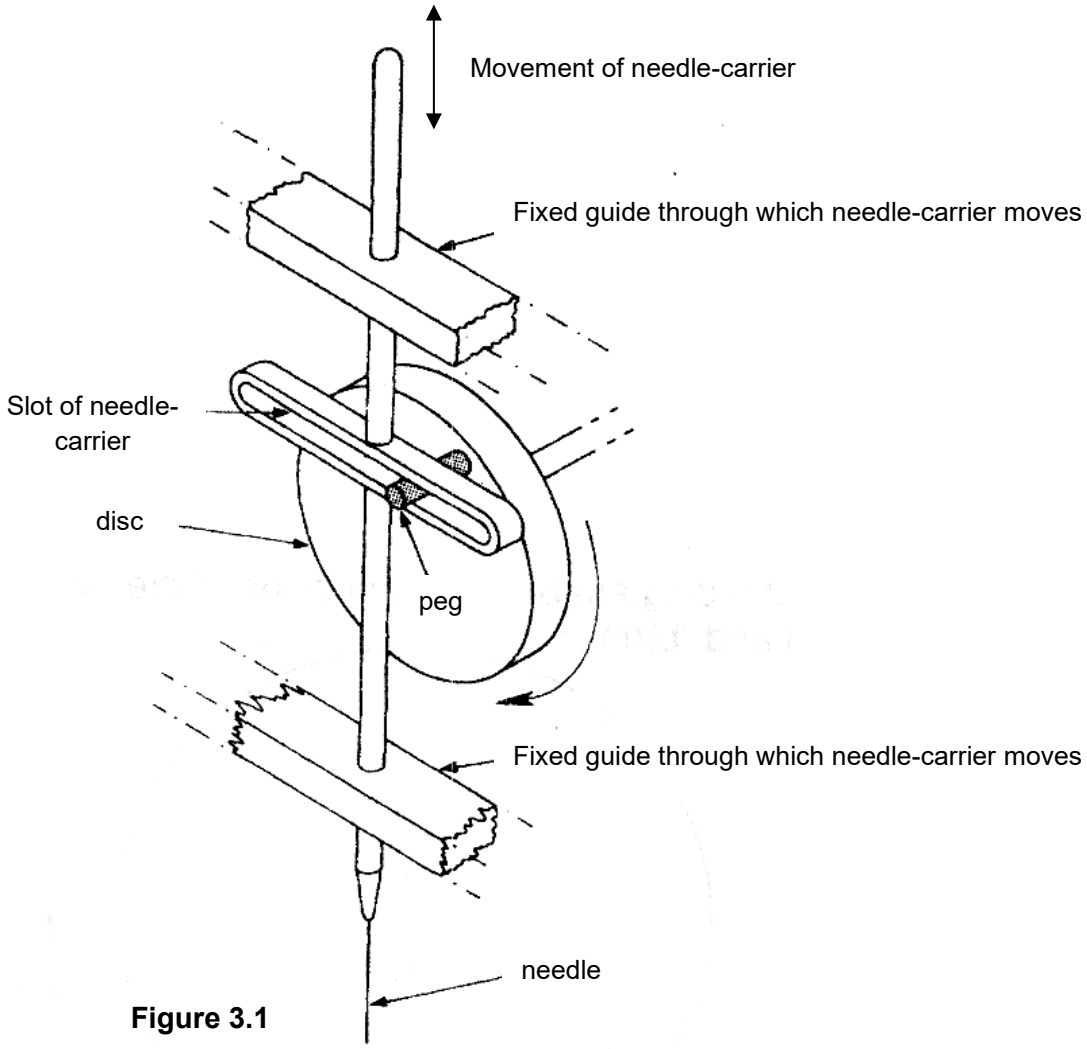



**Figure 3.1**

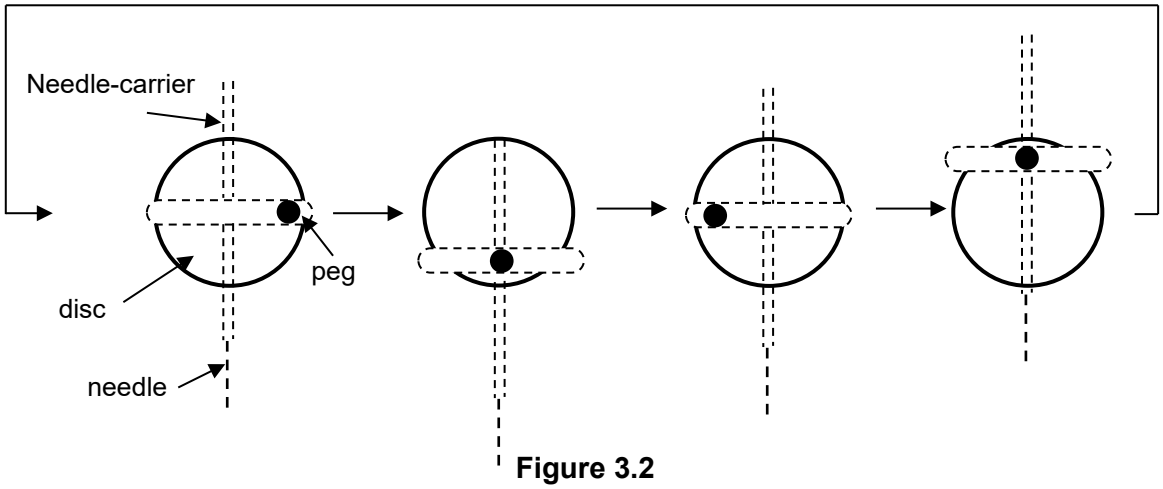



**Figure 3.2**

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**(a)** The machine makes 12 stitches per second.

Show that the angular speed of the peg's circular motion is 75 rad $s^{-1}$.

[1]

**(b)** From its highest to lowest point, the needle-carrier moves through a vertical distance of 32 mm.

**(i)** Calculate the maximum acceleration of the needle-carrier.

maximum acceleration = ..... m $s^{-2}$ [2]

**(ii)** Explain why the peg exerts a larger contact force on the slot of the needle-carrier when it is at its lowest position compared to when it is at its highest position.

.....

.....

.....

..... [2]

**(c)** The needle-carrier and needle have a total mass of 25 g.

Sketch in Figure 3.3 the variation of the kinetic energy of the needle-carrier and needle with time $t$ for two complete periods. The needle-carrier and needle are at the highest position at $t = 0$.

Indicate all relevant values on the axes.

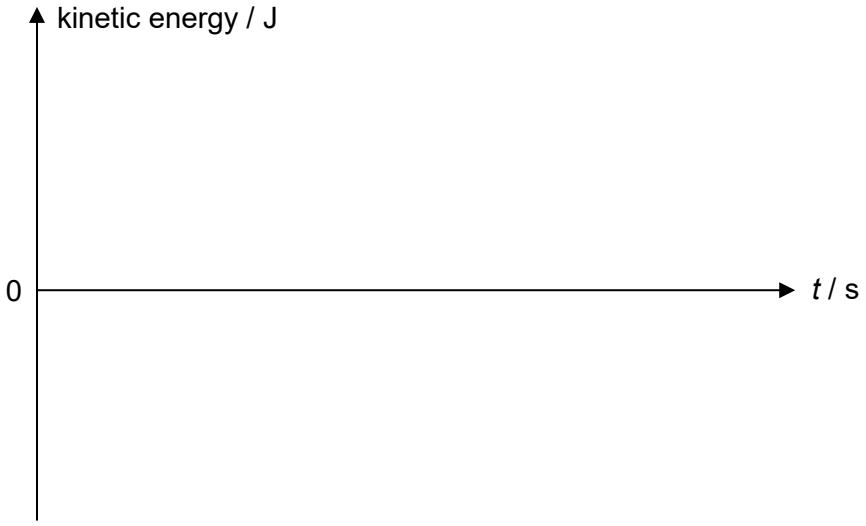



**Figure 3.3**

[3]

[Total: 8]

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**4** **(a)** With reference to molecular kinetic and potential energies, explain why the internal energy of the system increases when a wire is stretched within its elastic limit at constant temperature.

.....

.....

.....

..... [2]

**(b)** A fixed mass of an ideal gas is initially at a temperature of 17 °C.

The gas has a volume of 0.24 $m^3$ and a pressure of $1.2 \times 10^5$ Pa.

**(i)** Calculate the number of moles $n$ of the gas.

$n$ = ..... mol [1]

**(ii)** The gas undergoes three successive changes, as shown in Figure 4.1.

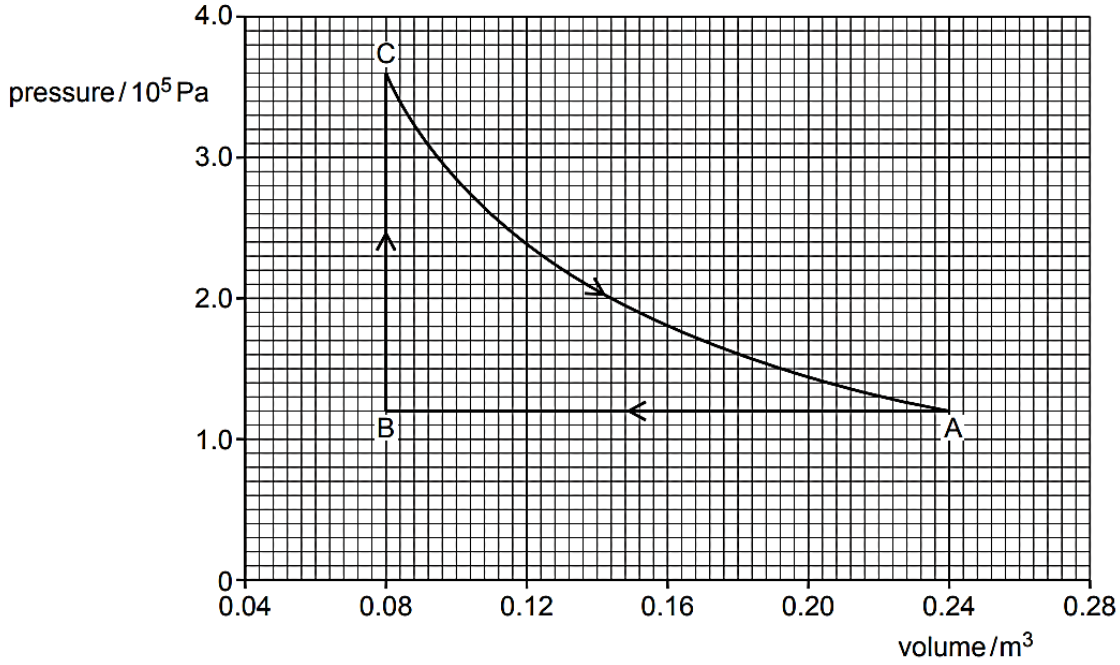



**Figure 4.1**

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, 31.6 kJ of work is done.

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Complete Table 4.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

| change | work done on gas / kJ | thermal energy supplied to gas / kJ | increase in internal energy of gas / kJ |
|---|---|---|---|
| AB | | –48.0 | |
| BC | | | |
| CA | | | |

**Table 4.1**

[3]

[Total: 6]

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**5** Student X constructs a circuit to demonstrate the effect of loading a battery by adding resistors in parallel across it. An ammeter of negligible resistance measures the total current drawn from the battery.

Resistors can be added into the circuit by closing switches **A**, **B**, **C**, **D**, **E** as shown in Figure 5.1.

The battery has an e.m.f. of 9.0 V and each of the resistor has a labelled value of 10 Ω.

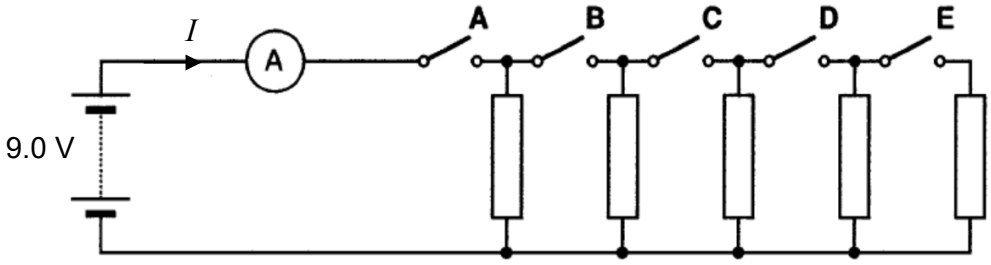



**Figure 5.1**

**(a)** Due to the way the resistors are made, there is a random variation of ± 5% in their value.

Student X assumes that the internal resistance of the battery is negligible.

Using this assumption and taking the e.m.f. of the battery to be constant, show that the possible variation in the value of the current $I$ when only switch **A** is closed is approximately 0.05 A.

[2]

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**(b)** Figure 5.2 shows Student X's predicted and actual results.

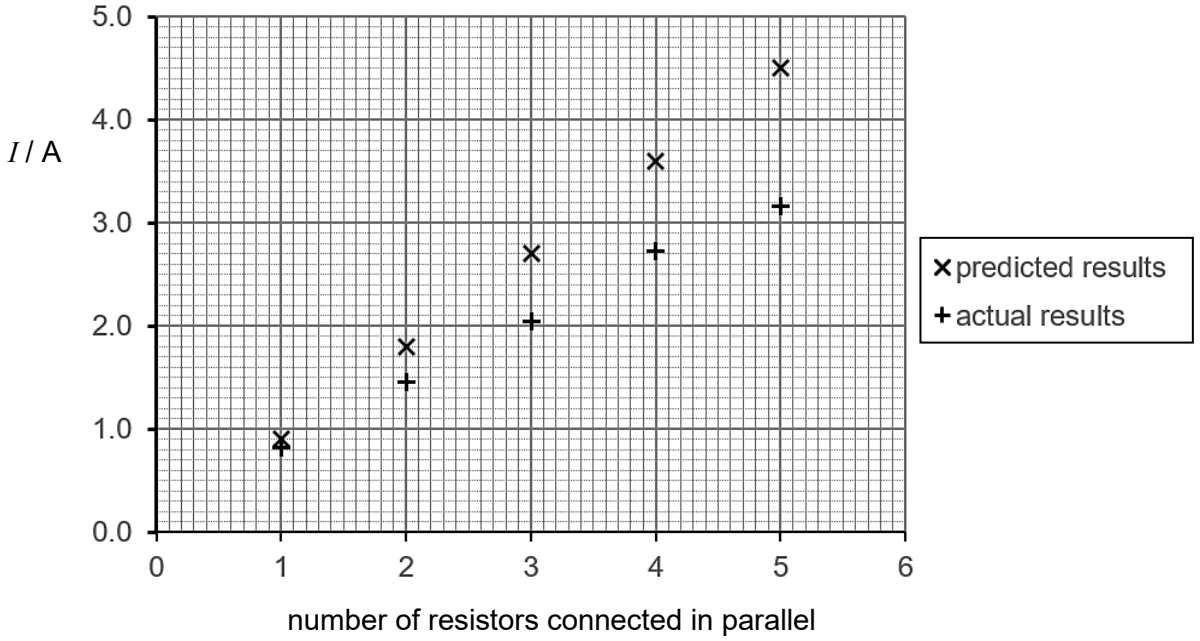



**Figure 5.2**

**(i)** Student X suggests that the difference between the two sets of results is due to the random variation (± 5%) of the value of the resistors.

Explain how the data in Figure 5.2 shows that Student X is **incorrect**.

.....

.....

..... [1]

**(ii)** Student Y then suggests that the increasing deviation between predicted and actual results are due to the internal resistance of the 9.0 V battery.

Explain how the data in Figure 5.2 supports Student Y's statement.

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..... [2]

**(c)** Student Y then plots a graph of $1/I$ against the resistance $R$ of the resistors in parallel to find the internal resistance of the battery. (Figure 5.3)

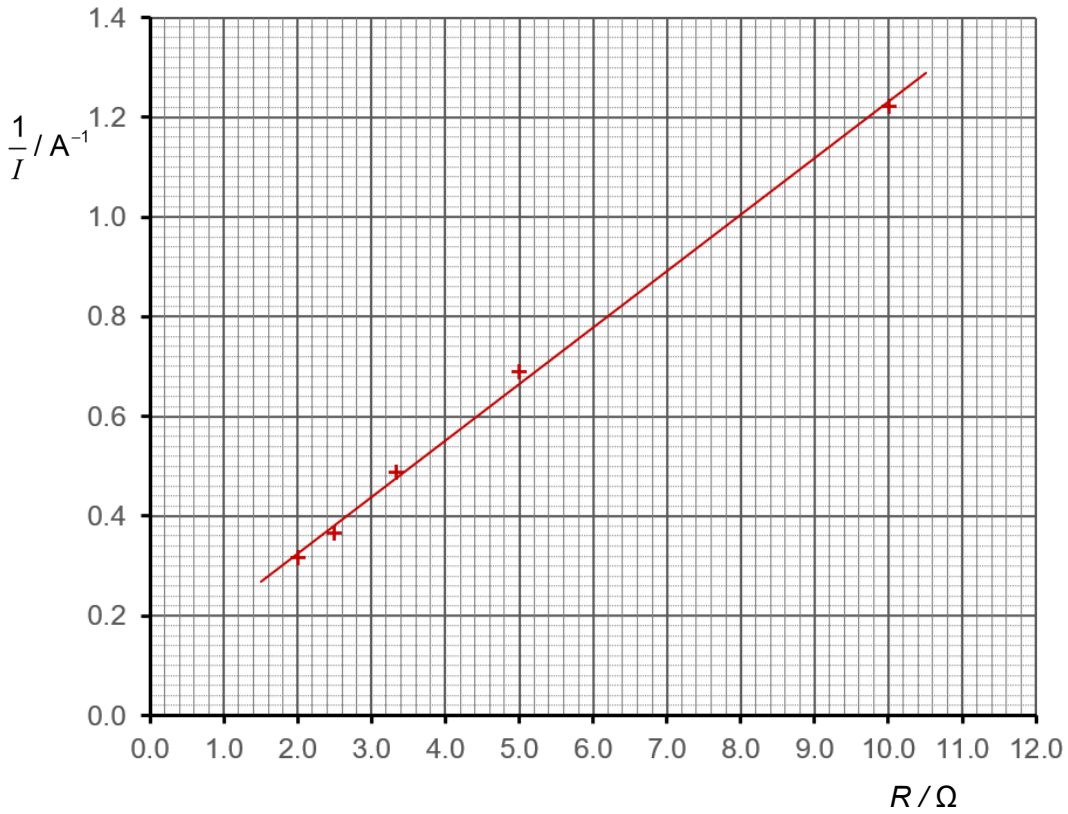



**Figure 5.3**

Use the graph to determine the internal resistance of the battery.

internal resistance = ..... Ω [3]

[Total: 8]

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**6** A solenoid is connected to an e.m.f source and a variable resistor as shown in Figure 6.1. A small coil connected to a galvanometer is placed close to the solenoid.

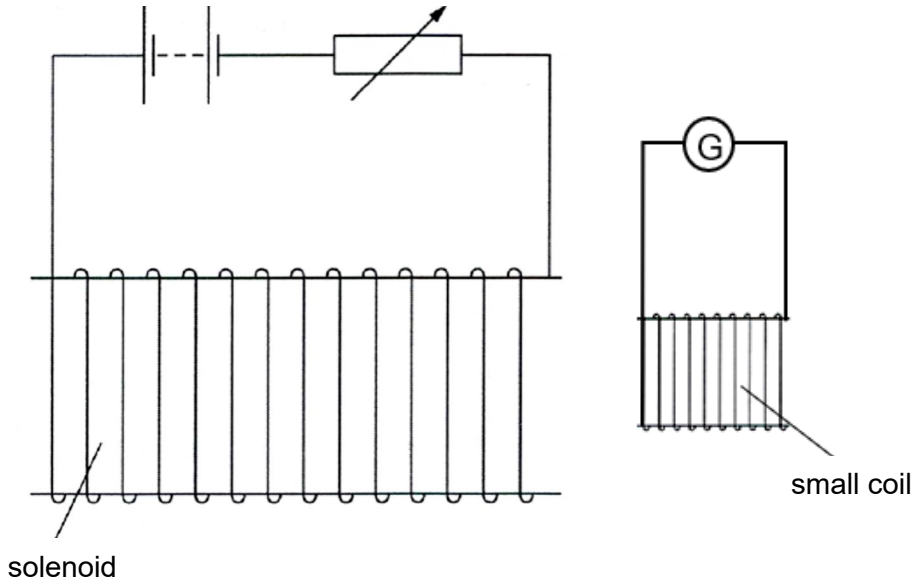



**Figure 6.1**

**(a)** The resistance of the variable resistor is slowly increased.

**(i)** State Faraday's law of electromagnetic induction and use it to explain why a reading is detected in the galvanometer.

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..... [2]

**(ii)** On Figure 6.1, mark the direction of the induced current passing through the small coil.

Explain the direction of the induced current.

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**(b)** The current in the solenoid is now made to vary as shown in Figure 6.2.

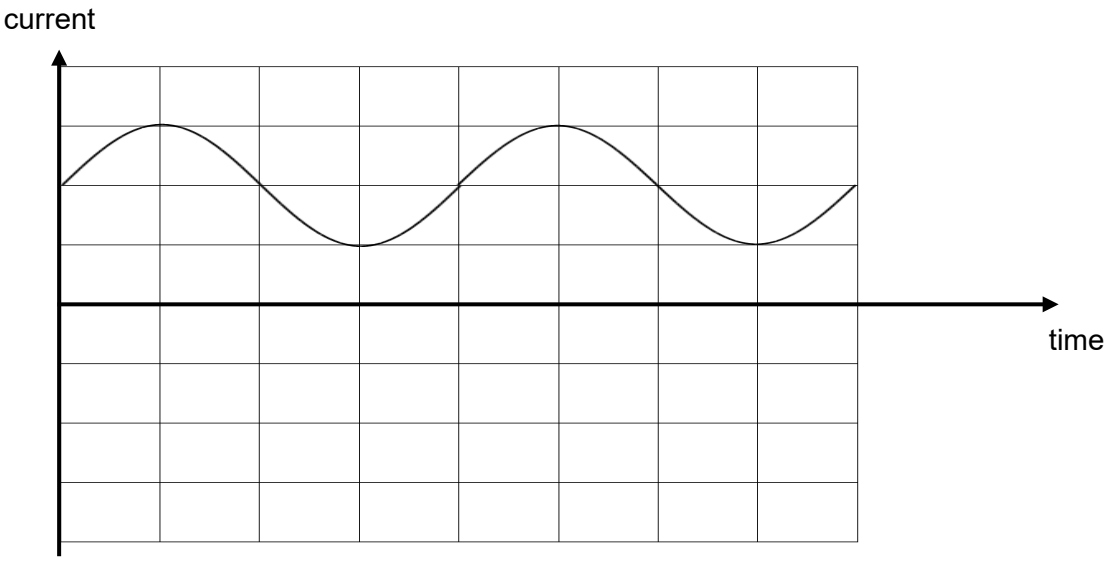



**Figure 6.2**

**(i)** On Figure 6.3, sketch the variation with time of the e.m.f. induced in the small coil. Label it as **P**. The time scale is the same. [1]

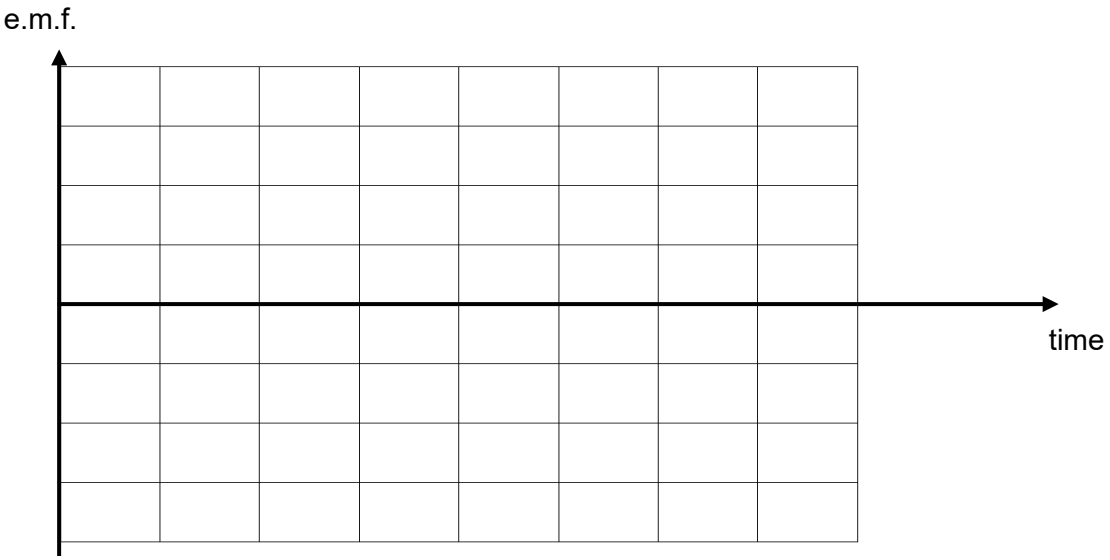



**Figure 6.3**

**(ii)** The frequency of the current in the solenoid is halved.

On Figure 6.3, sketch the new variation of the induced e.m.f. in the small coil and label it as **Q**.

[2]

[Total: 7]

**7** **(a)** Explain how the photoelectric effect provides evidence that supports the particulate nature of electromagnetic radiation.

.....

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..... [2]

**(b)** The SARS-CoV-2 virus measures 140 nm in diameter.

Explain why it can be observed under an electron microscope but not an optical microscope that uses visible light.

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**(c)** A solar sail uses light to propel a spacecraft through space.

Explain how massless photons exert a force on a solar sail.

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**8** Figure 8.1 shows the variation with nucleon number $A$ of the binding energy per nucleon $B_E$ of a nucleus.

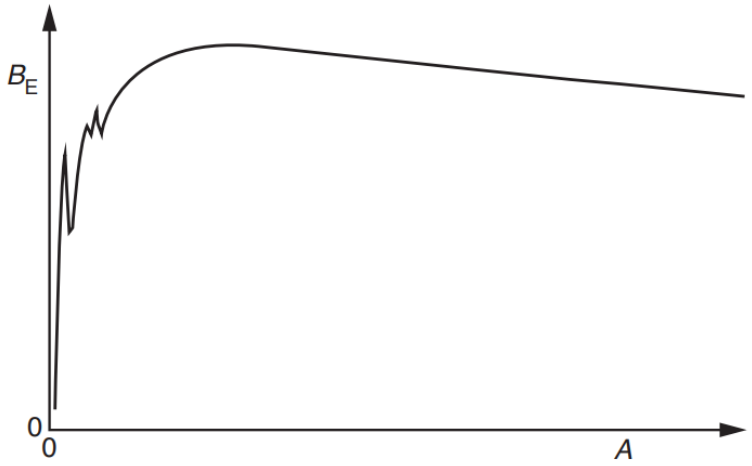



**Figure 8.1**

When a nucleus of uranium-235 ($^{235}_{92}\text{U}$) absorbs a neutron in a nuclear reactor, it undergoes fission to form various products and release further neutrons.

In one nuclear reaction involving the fission of uranium-235, barium-141 ($^{141}_{56}\text{Ba}$) and krypton-92 ($^{92}_{36}\text{Kr}$) are the products. Three neutrons are also released when these products are formed, and energy is released.

The equation for this reaction is shown:

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^{1}_{0}\text{n}$$

**(a)** Explain, with reference to the line in Figure 8.1, why this reaction results in the release of energy.

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..... [2]

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**(b)** Uranium-235 has a binding energy per nucleon of 7.5909 MeV.

Data for some masses are given in Table 8.1

| | mass/u |
|---|---|
| proton | 1.00814 |
| neutron | 1.00898 |
| barium-141 | 140.91441 |
| krypton-92 | 91.92615 |

**Table 8.1**

**(i)** Show, to 5 decimal places, that the mass of a uranium-235 nucleus is 235.12259u.

[2]

**(ii)** Calculate the energy released in this fission reaction.

energy = ..... J [2]

**(iii)** The total kinetic energy of the products for the reaction is found to be more than the answer in **(b)(ii)**.

Suggest a reason for this difference.

.....

.....

..... [1]

[Total: 7]

## Section B

Answer **one** question from this section in the spaces provided.

**9** **(a)** State the principle of conservation of momentum.

.....

.....

..... [2]

**(b)** Two frictionless trolleys A and B with attached identical springs approach each other along a horizontal line, as illustrated in Figure 9.1.

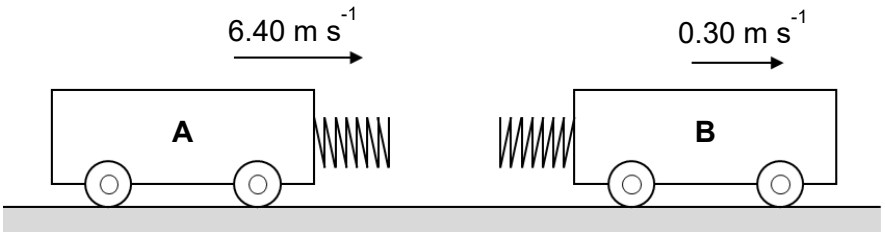



**Figure 9.1**

Trolley A has mass 1.2 kg and a velocity of 6.40 m $s^{-1}$. Trolley B has mass 5.2 kg and a velocity of 0.30 m $s^{-1}$.

The two trolleys collide elastically. After collision, trolley A reverses its direction of motion.

**(i)** Show that the speed of trolley B after the collision is 2.6 m $s^{-1}$.

[3]

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**(ii)** Figure 9.2 shows the velocity-time graph of trolley B.

Sketch also on Figure 9.2, the velocity-time graph for trolley A and label it as **A**.

Graph of velocity against time

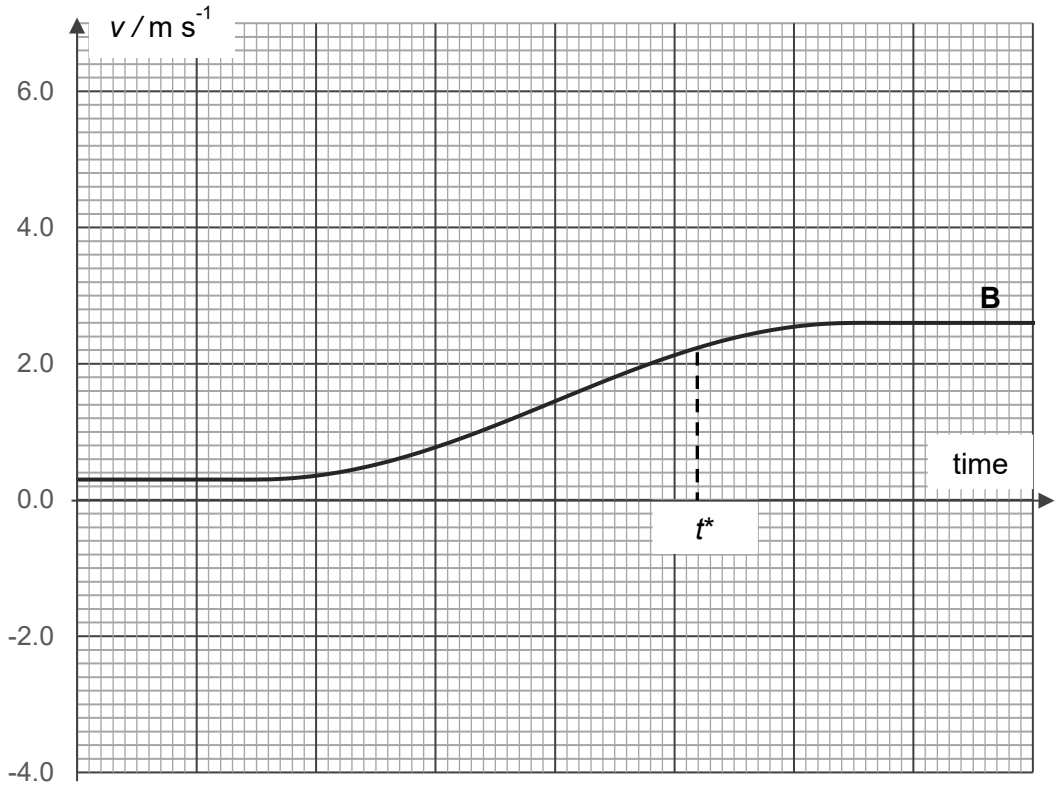



**Figure 9.2**

[3]

**(iii)** Explain how the contact force on trolley B by trolley A at time *t** can be determined from Figure 9.2.

.....

.....

..... [2]

**(iv)** Describe the change in the total kinetic energy of the system of trolleys A and B throughout the collision.

.....

.....

..... [2]

**(v)** Calculate the maximum elastic potential energy stored in *each* spring.

maximum elastic potential energy = ..... J [3]

**(c)** After the collision of the two trolleys, trolley B moves to the edge of the platform and falls through a vertical height of 10.0 m before being brought to rest in a duration of 0.100 s, without rebounding.

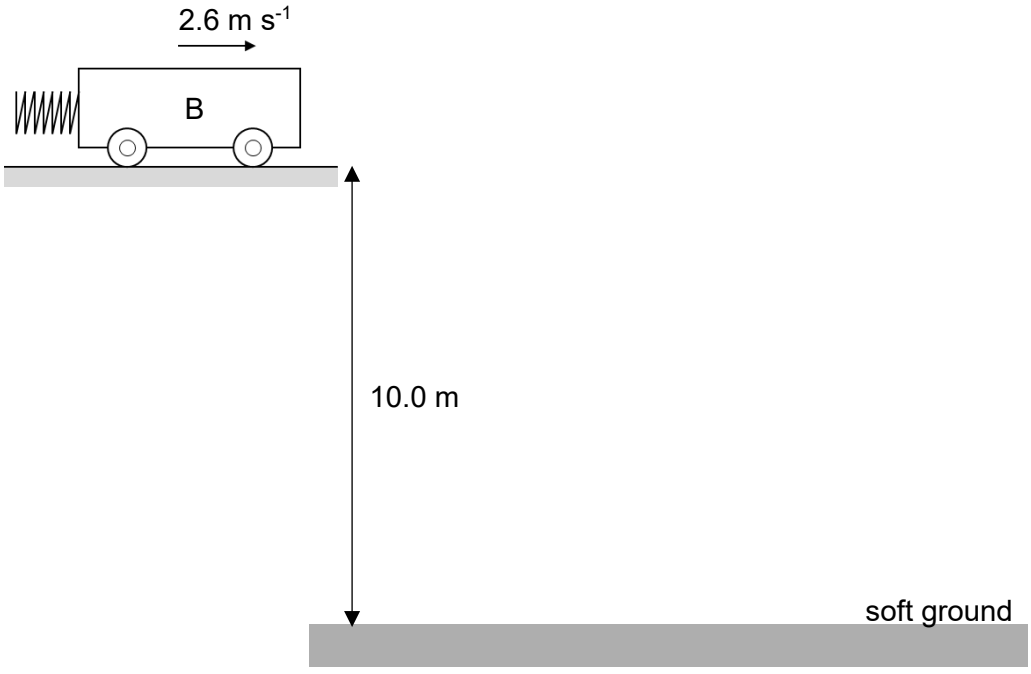



**Figure 9.3** (not to scale)

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**(i)** Determine the average vertical contact force acting on the trolley by the soft ground.

force = …… N [4]

**(ii)** Explain what happens to the impulse on the trolley if it were to land on hard ground and rebounded instead.

.....

..... [1]

[Total: 20]

**10** **(a)** State what is meant by coherent light waves.

.....

..... [1]

**(b)** Coherent light of wavelength of 633 nm is incident normally on a pair of narrow parallel slits separated by a distance of 0.250 mm as shown in Figure 10.1.

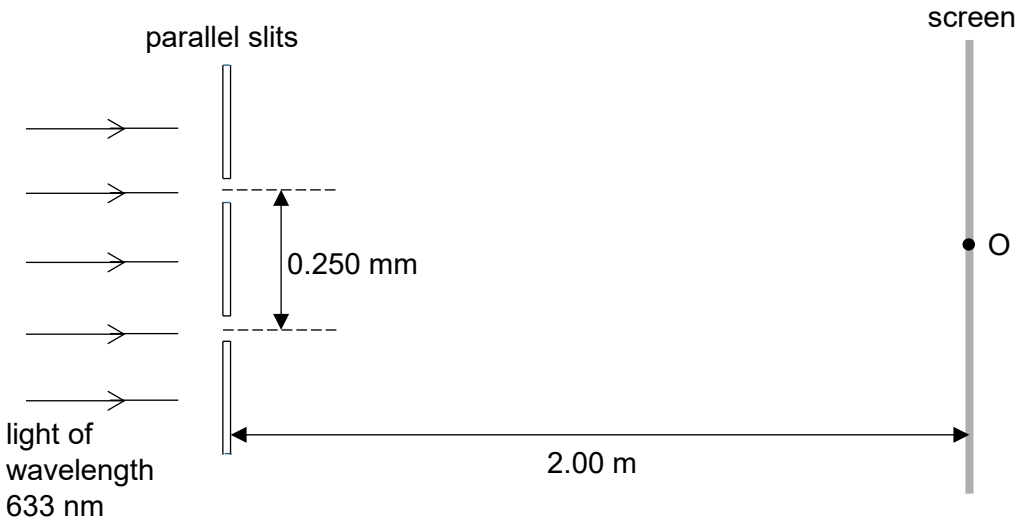



**Figure 10.1** (not to scale)

An interference pattern of bright and dark fringes is seen on a screen placed parallel to the plane of the slits. The screen is 2.00 m away from the slits. Point O is equidistant from both slits.

**(i)** Explain how the bright and dark fringes are formed on the screen.

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..... [2]

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**(ii)** Calculate the distance between the central bright fringe and the second-order dark fringe on the screen.

distance = ..... mm [2]

**(iii)** Explain why the maximum observed at O is of a greater intensity than any other observable maximum.

.....

.....

.....

.....

..... [2]

**(c)** The setup is modified by blocking one of the slits, leaving a single slit of width 0.080 mm exposed to the original light source of single wavelength of 633 nm as shown in Figure 10.2.

A diffraction pattern of the single slit is formed on the screen centred at point P.

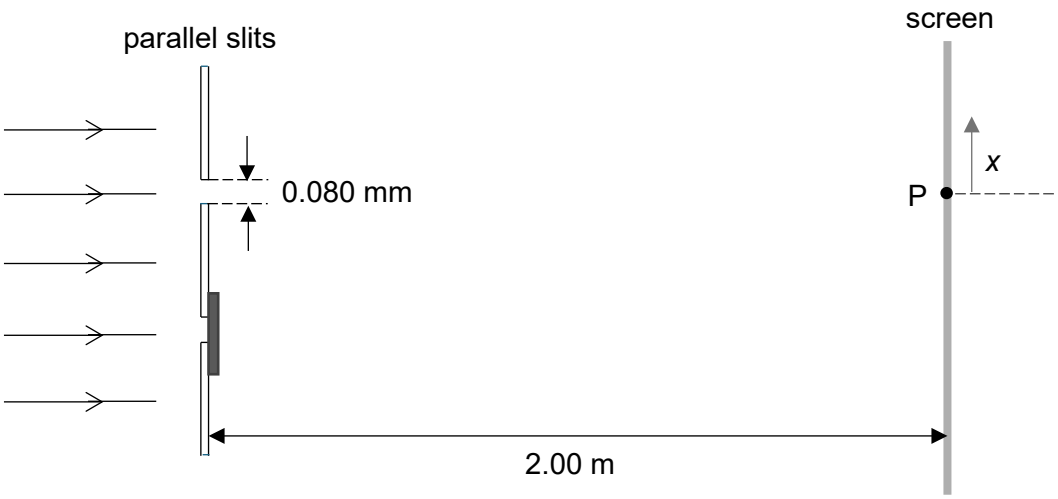



**Figure 10.2** (not to scale)

**(i)** Show that the width of the central fringe on the screen is approximately 31.7 mm.

[2]

**(ii)** Part of the diffraction pattern showing the variation with displacement $x$ from P of the intensity of light is shown in Figure 10.3.

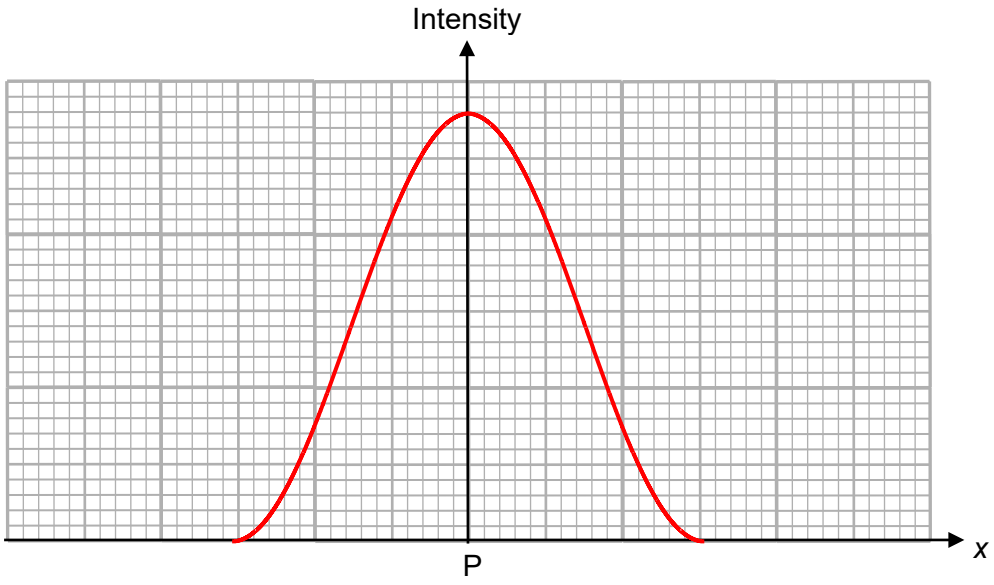



**Figure 10.3**

The slit width is reduced by half. Sketch on Figure 10.3 the new diffraction pattern. [2]

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**(iii)** The single slit is now used as an aperture for a simple optical imaging system.

Two small LED indicators are spaced 5.0 mm apart as shown in Figure 10.4. The LED indicators emit light of wavelength 633 nm.

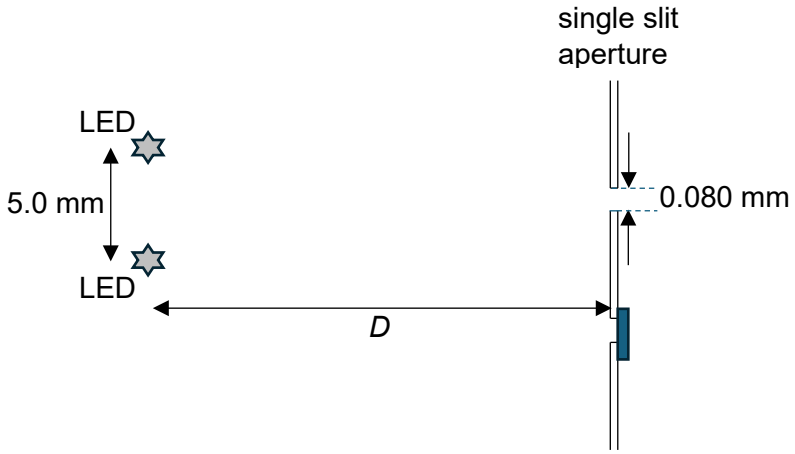



**Figure 10.4** (not to scale)

Calculate the maximum distance *D* such that the images from the LED indicators can be just resolved.

*D* = ..... m [3]

**(d)** A parallel beam of light of wavelength 633 nm is incident normally onto a 6.00 m screen. A bright spot is observed at the centre of the screen.

A diffraction grating is inserted normally into the path of the light as shown in Figure 10.5. The grating has 500 lines per millimetre.

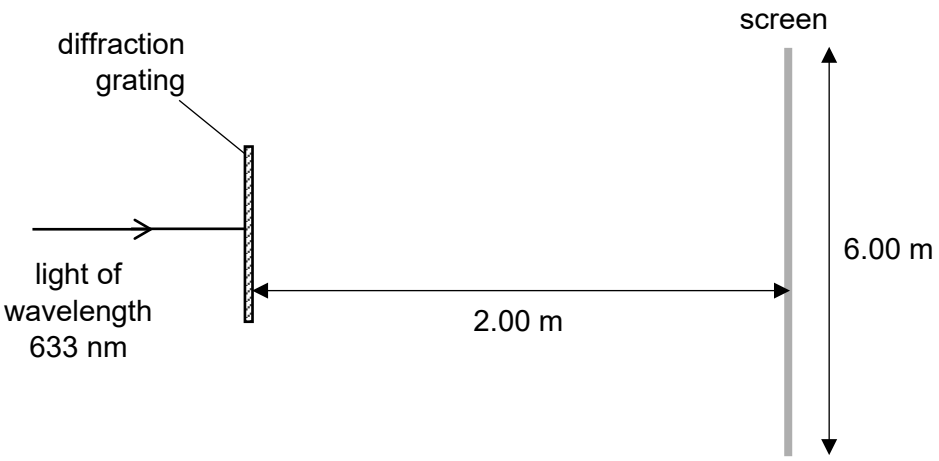



**Figure 10.5** (not to scale)

**(i)** Determine the maximum number of bright lines that can be observed on the screen.

maximum number = …….. [4]

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**(ii)** The incident beam is now changed to a source of white light.

State and explain how the appearance of the first-order spectrum differs from the central maximum.

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..... [2]

[Total: 20]

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